

Java FSD - Day 8 Summary

Pattern Printing Using For Loop

- Day 8 focused on solving different pattern printing programs using nested for loops.
- Pattern questions help improve logical thinking and understanding of loop execution.

Right-Aligned Triangle Pattern

- This pattern prints stars in a right-aligned triangle shape.
- One loop is used to print spaces and another loop is used to print stars.
- The number of stars increases in each row.

Pyramid Star Pattern

- A pyramid pattern is created using spaces and stars.
- The outer loop controls rows, while inner loops control spaces and stars.
- The formula $(2 \times i - 1)$ is commonly used to print the stars.

Additional Pattern Questions

- Inverted Pyramid Star Pattern
- Square Star Pattern
- Number Pyramid (1 to n)
- Repeated Number Triangle
- Reverse Number Triangle
- Diamond Star Pattern

Important Points

- Nested loops are the main concept behind pattern printing.
- Outer loop controls rows and inner loop controls columns.
- Pattern programs are commonly asked in interviews and coding assessments.

Revision Note

Day 8 mainly focused on understanding nested for loops and creating different star and number patterns using Java.

Reference Images

Pattern Printing Questions Using for Loop

3. Write a Java program to print a right-aligned triangle star pattern.

```
public class Pattern3 {  
    public static void main(String[] args) {  
        for (int i = 1; i <= 5; i++) {  
            for (int j = 5; j > i; j--) {  
                System.out.print(" ");  
            }  
            for (int k = 1; k <= i; k++) {  
                System.out.print("* ");  
            }  
            System.out.println();  
        }  
    }  
}
```



Pattern Printing Questions Using for Loop

4. Write a Java program to print a pyramid star pattern.

```
public class Pattern4 {  
    public static void main(String[] args) {  
        int rows = 5;  
        for (int i = 1; i <= rows; i++) {  
            for (int j = rows; j > i; j--) {  
                System.out.print(" ");  
            }  
            for (int k = 1; k <= (2 * i - 1); k++) {  
                System.out.print("*");  
            }  
            System.out.println();  
        }  
    }  
}
```



Pattern Printing Questions Using for Loop

5. Write a Java program to print an inverted pyramid star pattern.
6. Write a Java program to print a square pattern of stars.
7. Write a Java program to print a number pyramid (1 to n).
8. Write a Java program to print a repeated number triangle (same number per row).
9. Write a Java program to print a reverse number triangle.
10. Write a Java program to print a diamond star pattern.